untitled

File Edit Format Run Options Window Help

```
from collections import deque

def bfs(graph, start):
    visited = {start}
    queue = deque([start])
    order = []
    while queue:
        node = queue.popleft()
        order.append(node)
        for neighbour in graph[node]:
            if neighbour not in visited:
                visited.add(neighbour)
                queue.append(neighbour)
    return order

def bfs_shortest_path(graph, start, goal):
    parent = {start: None}
    queue = deque([start])
    while queue:
        node = queue.popleft()
        if node == goal:
            path = []
            while node is not None:
                path.append(node)
                node = parent[node]
            return path[::-1]
        for neighbour in graph[node]:
            if neighbour not in parent:
                parent[neighbour] = node
                queue.append(neighbour)
    return None

if __name__ == "__main__":
    graph = {
        "A": ["B", "C"],
        "B": ["D", "E"],
        "C": ["F", "G"],
        "D": [], "E": [], "F": [], "G": [],
    }
    print("BFS order from A     :", " -> ".join(bfs(graph, "A")))
    print("Shortest path A to G :", " -> ".join(bfs_shortest_path(graph, "A", "G")))
    g2 = {1: [2, 3], 2: [1, 4], 3: [1, 4], 4: [2, 3, 5], 5: [4]}
    print("BFS order from 1 (cyclic graph):", bfs(g2, 1))
```

Ln: 43 Col: 57

26°C Light rain

ENG IN 09:32 25-10-2026

```
==========================================================
BFS order from A     : A -> B -> C -> D -> E -> F -> G
Shortest path A to G : A -> C -> G
BFS order from 1 (cyclic graph): [1, 2, 3, 4, 5]
```